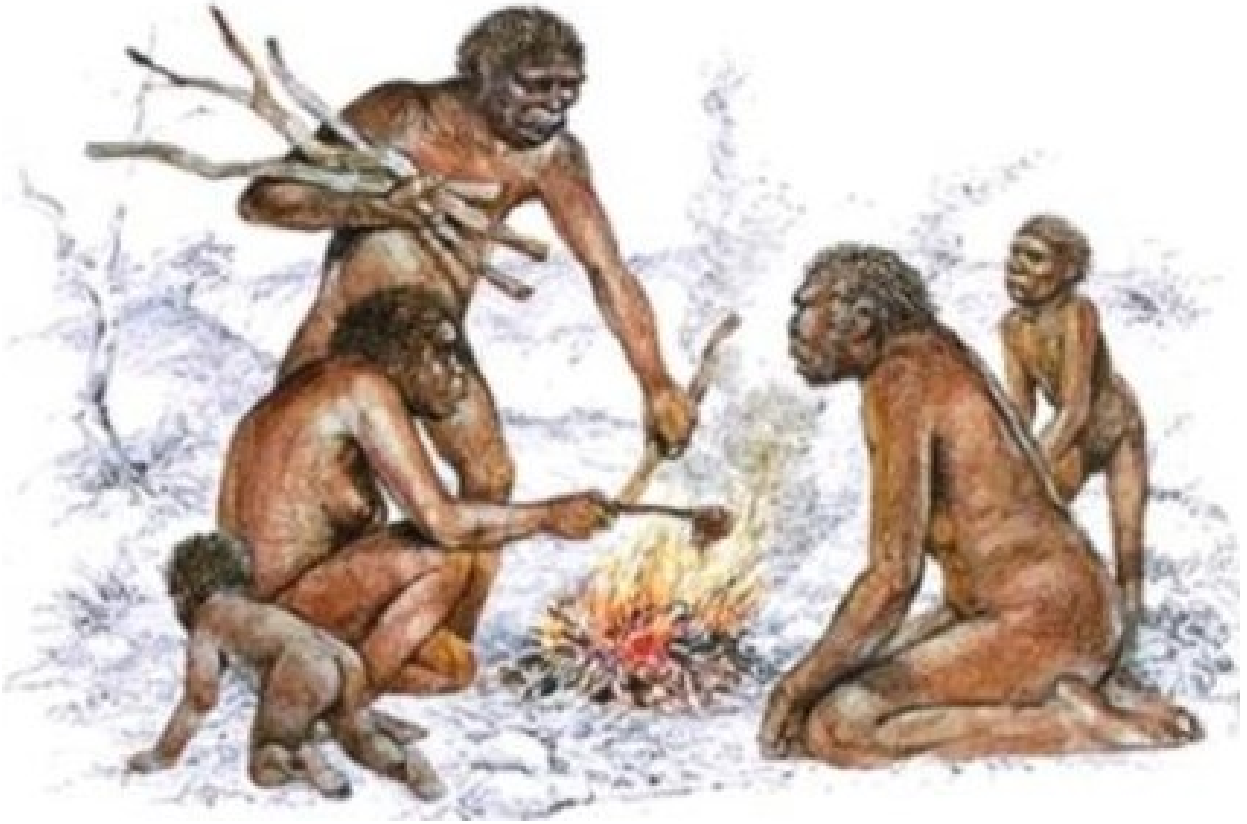
Human Ancestors: *Homo habilis, Homo erectus*



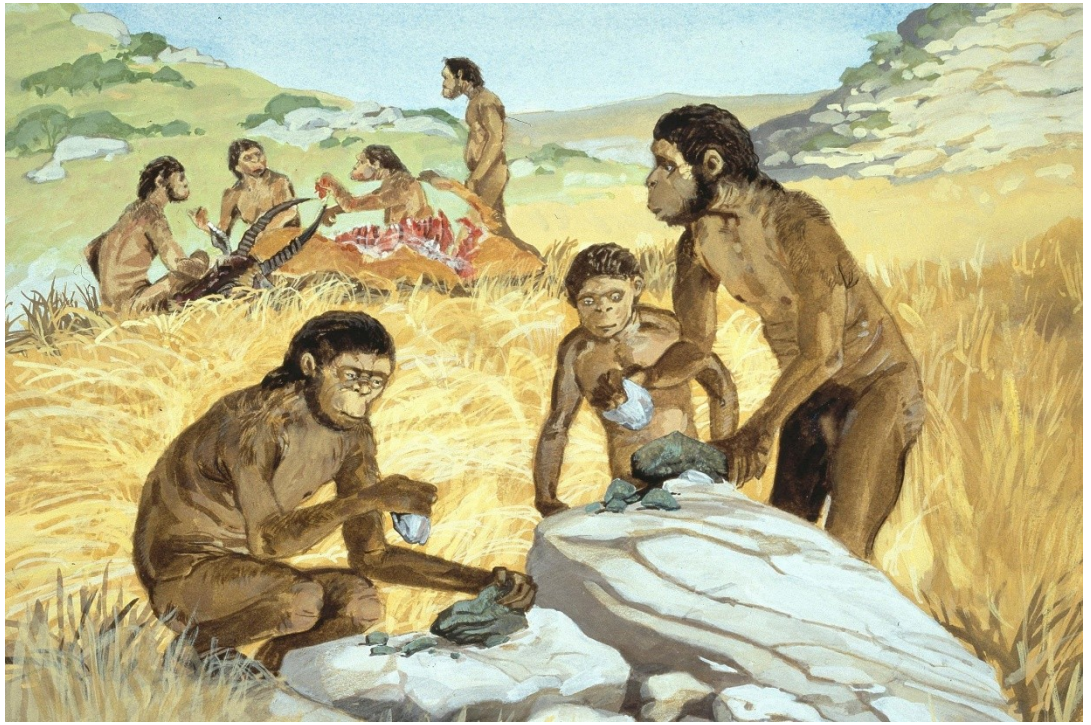
Habitat and Behaviour

Fossil Record

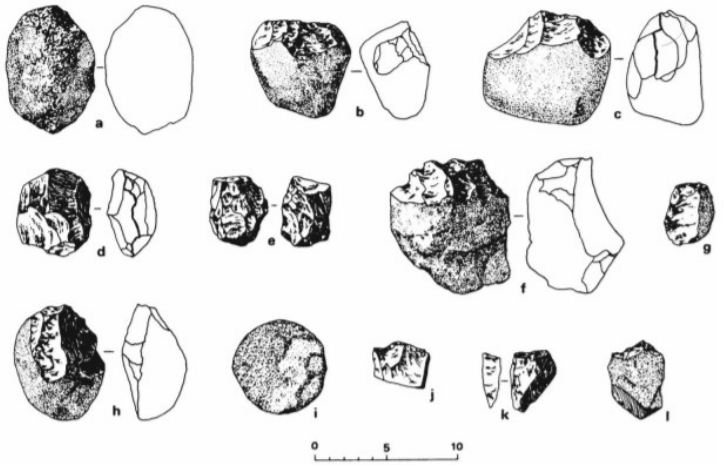
Anatomical Features

Homo habilis: “Handy Man” – Overview

- Likely to have evolved from *Australopithecus*.
- Lived 2.4 million to 1.4 million years ago in East Africa.
- May have been a direct ancestor of humans, or on a different “branch” of the evolutionary tree.
- Bipedal
- Lived in family groups.
- Probably spent daylight hours on the ground, may have slept in trees for safety.
- Crafted stone tools “Oldowan choppers” – may have been the first to do so.
- Evidence of animal bones being chopped up to get to marrow.
- Tooth enamel studies suggest varied diet including meat.







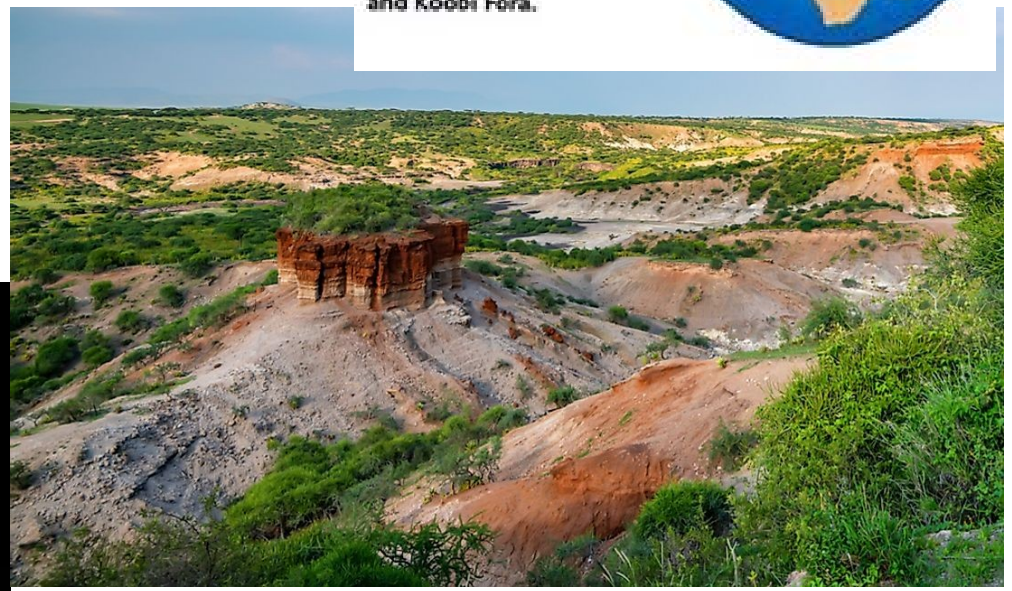
Learning Aim: Describe the habitat, lifestyle, time of existence and distribution of Homo habilis

Homo habilis – Fossil Record

- First find in 1964 – Dr Louis Leakey and team
  - Found jaw, two cranial fragments and other remains
  - Olduvai Gorge – east Africa
  - Dated at 1.75 million years BP
  - Named *"Homo habilis"* – "handy man" – tool making
  - Larger brain and smaller teeth than *Australopithecus*, with slightly more erect posture.
- Further finds since then
- Can be hard to classify between *H. habilis and H. erectus*



Map of Africa showing the locations of Olduvai Gorge and Koobi Fora.





Learning Aim: Describe the location of major fossil finds for Homo habilis.

Homo habilis – Anatomy



- Height – females 110cm, males 130cm
- Cranial capacity around 610cc.
- Rounder skull
- Small brow ridge
- Central foramen magnum
- Moderate prognathism
  - Less jutting than *Australopithecus*, more than later humans
- Increasingly parabolic dental arcade
- Curved finger bones – powerful grip
- Able to form precision grip
- Fossils of feet show structure similar to modern humans





Learning Aim: Describe anatomical features of Homo habilis.

Homo habilis



Relationship to humans:

- A possible direct ancestor of humans, coexisted with other hominids

Time Period:

- 1.5-2.5 mya
- Pleiocene/Pleistocene

Distribution:

- East Africa, esp Olduvai Gorge



Climate:

- Spreading grasslands and savannah
- Selection pressure for bipedalism and adaptability.

Fossil Record and what it tells us:

- Variety of fossil fragments found in various East African locations, including Olduvai Gorge:
  - Foot fragments—aligned toe, arch similar to modern humans—bipedal.
  - Skull—larger cranial capacity than *Australopithecus,* smaller brow ridge.
  - Skull fragments—prognathic jaw but less so than *Australopithecus,* foramen magnum at base—bipedal
  - Hand fragments— large hands, thumb similar to modern chimps
  - Arm fragments—longer arms than modern humans
  - Jaw fragments—dental arcade rounder than *Australopithecus,* teeth smaller.



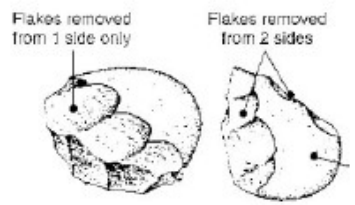


- Stone tools found, same age as fossils— "Oldowan" Tool Culture

Cultural Evolution:

Tool Use

- "Olduwan" Tool Culture (may also be *Australopithecine)*
  - Crudely flaked river pebbles
  - Crafted into choppers
  - Large chips—not finely crafted



Fire Use

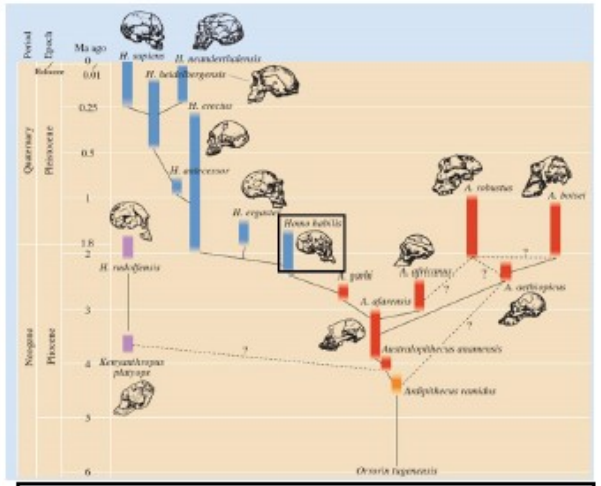
- No evidence of fire use

Language, Art, Abstract Thought

- No evidence of complex Language, art or abstract thought

Built Structures

- Made shelters from sticks and vegetation
- No evidence of clothing



Lifestyle:

- In grassland
- Bipedal gait
- Foraging and scavenging
- Family groups
- Made shelters and used tools

Diet and Dentition:

- Mostly fruit, leaves, vegetation—some meat.
- Teeth show rounded dental arcade:
  - Tough enamel—cope with vegetation
  - Smaller teeth than *Australopithecus,* but larger incisors than humans

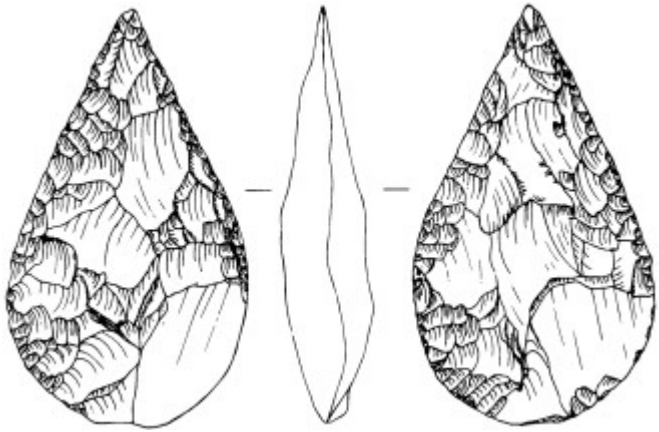


Homo erectus: "Upright Man" – Overview



- Likely to have evolved from *Australopithecus*.
- Lived around 1.9 mya to 100 000 years before present.
  - Evidence of adaptation and evolution within the species in this time.
- Likely to be a direct ancestor of humans.
- Can be difficult to decide if fossils are *erectus* or *habilis*.
- Originally evolved in East Africa but migrated into Asia.
- African *Homo erectus* sometimes called *Homo ergaster*.
- Lived on ground, able to walk/run long distances
- Evidence of fire use – hearths for cooking, warmth, and to deter predators
- Made *Acheulian stone* tools.





Learning Aim: Describe the habitat, lifestyle, time of existence and distribution of Homo erectus.

Homo erectus – Fossil Record

- First fossils discovered in late 1800s – *"Java Man"* - Dubois
- 1927: *"Peking Man"* – *H. erectus* – Dr Davidson Black
  - Found in limestone cave south of Beijing
  - Brain larger, more modern features
  - Dental arcade shorter and rounded
  - Chin beginning to form, short, rounded jaw
  - Teeth very similar to modern humans – similar diet
  - Stone tools and hearth also found
  - Age 500 000 years ago – more recent than Java Man
- Footprints found in 2009
  - At Ileret in Kenya
  - 1.5 million years old
  - Indicate evolutionary origins in Africa
  - Human-like foot anatomy – big toe parallel, toes short, foot arch
  - *Homo ergaster* (often reclassified as African *Homo erectus*)



Learning Aim: Briefly outline aspects of the fossil record for Homo erectus.

Table 19.2 Some of the discoveries of early *Homo*

Common name	Scientific name	Age (years BP)	Year of discovery	Site
1470	*Homo rudolfensis*	1.9 million	1972	Koobi Fora (Africa)
	Homo erectus	1.8 million	1936	Java (Indonesia)
1813	*Homo habilis*	1.8 million	1973	Koobi Fora (Africa)
Handy man	*Homo habilis*	1.75 million	1960	Olduvai (Africa)
Turkana boy	*Homo ergaster*	1.6 million	1984	Nariokotome (Africa)
	Homo ergaster	1.6 million	2000	Ileret (Africa)
	Homo habilis	1.44 million	2000	Ileret (Africa)
	Homo erectus	1.4 million	1993	Dmanisi (Georgia)
	Homo erectus	1.0 to 1.4 million	1993	Java (Indonesia)
	Homo antecessor	780 000	1994	Gran Dolina (Spain)
Java man	*Homo erectus*	500 000 to 700 000	1891	Java (Indonesia)
Arago 21	*Homo heidelbergensis*	450 000	1971	Tautavel (France)
	Homo erectus	400 000 to 500 000	1966	Terra Amata (France)
	Homo erectus	300 000 to 500 000	1965	Vertésszöllös (Hungary)
Peking man	*Homo erectus*	250 000 to 350 000	1927	Zhoukoudian (China)

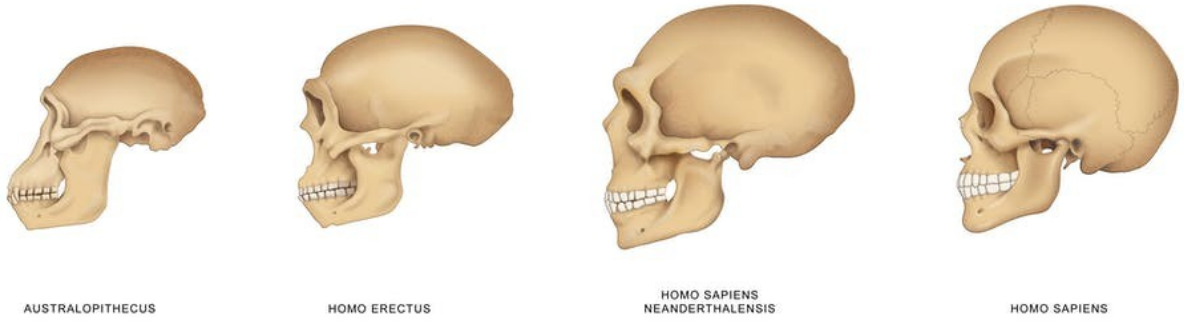


Learning Aim: Briefly outline aspects of the fossil record for Homo erectus.

Homo erectus – Anatomy

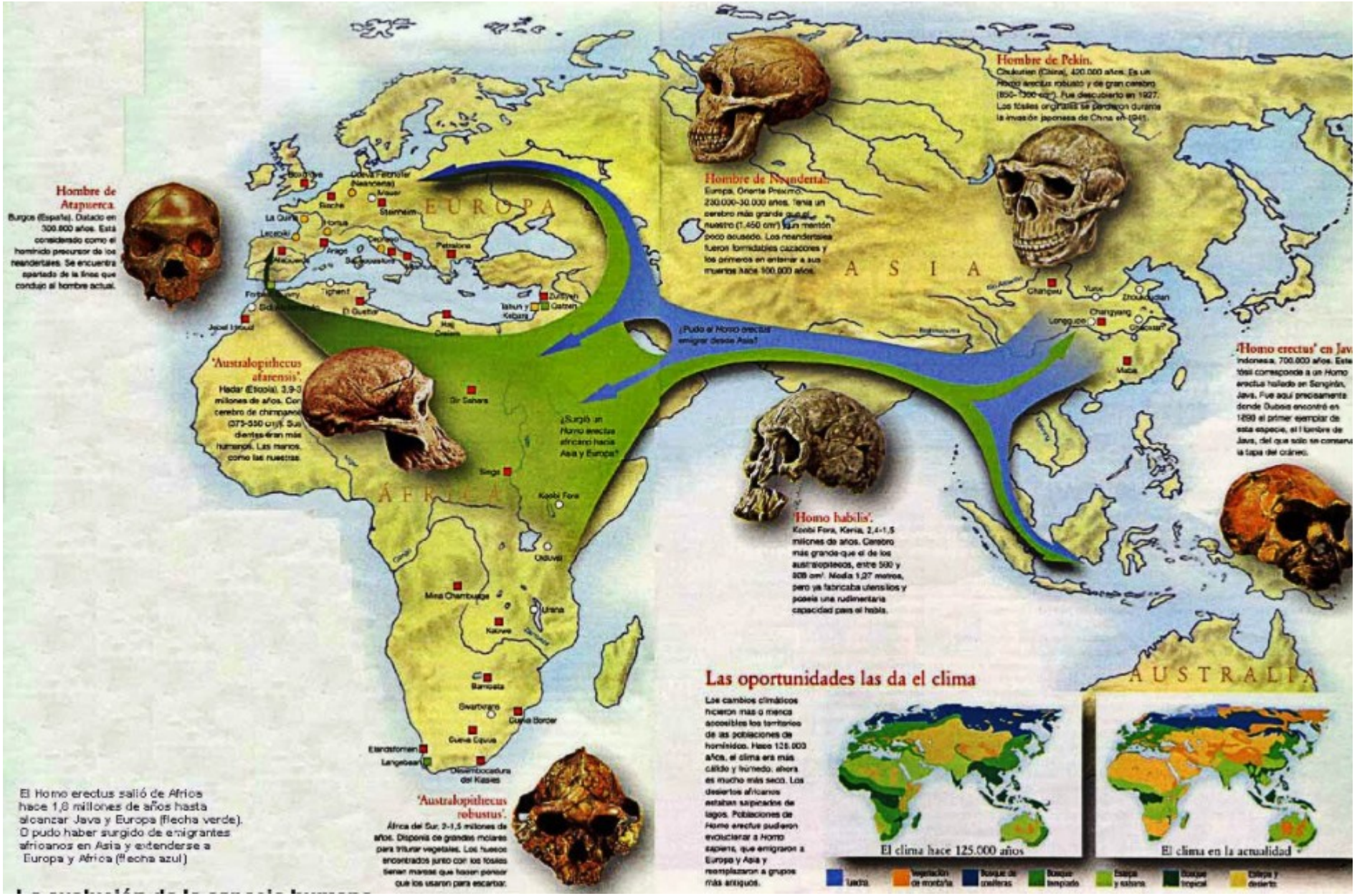
- Varied height 145-185 cm (similar to humans)
- Stocky body, thick bones
- Cranial capacity 1050cc
  - Much larger than previous hominins
- Low, sloping forehead
- Defined brow ridges
- Dental arcade increasingly parabolic, reduced molar size
- Less prognathism
- Teeth very similar to modern humans – varied diet





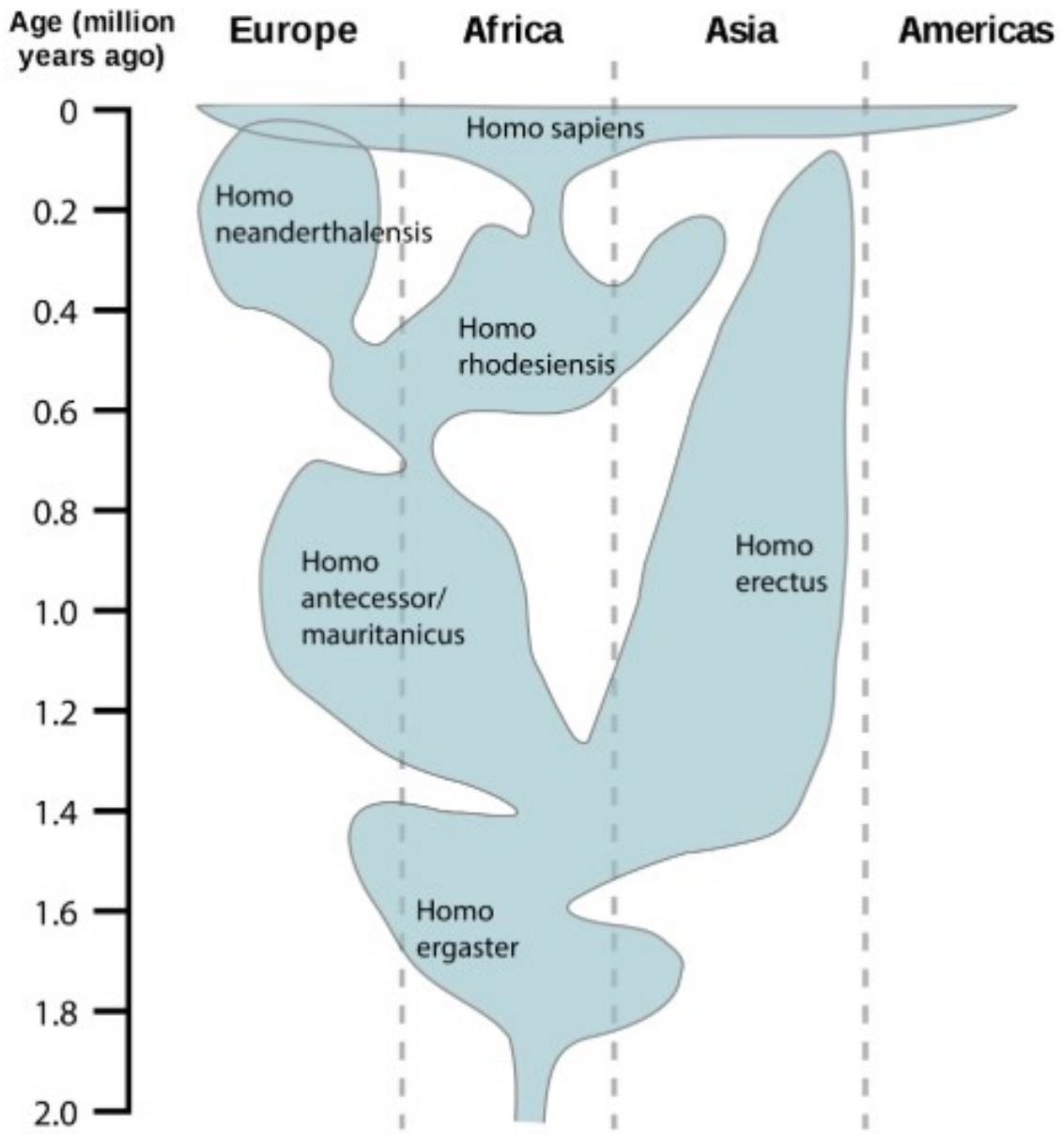


Learning Aim: Describe aspects of the anatomy of H.erectus and describe how this differs from earlier and later hominins.



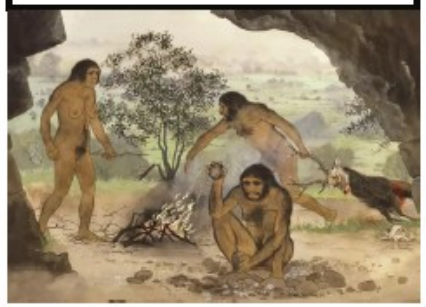
Learning Aim: Describe migration patterns and distribution for Homo erectus.

Migration patterns over time of *Homo* species



Learning Aim: Describe migration patterns and distribution for Homo erectus.

Homo erectus



Relationship to humans:

- A likely direct ancestor of humans, coexisted with other hominids, evolved into several different branches.
- Early *H.erectus* in Africa sometimes classified as *H.ergaster*

Time Period:

- 150 000 — 2 mya
- Pleistocene

Distribution:

- East Africa, migrating out to Asia and Europe



Climate:

- Variety of climates across range
- Selection pressure for bipedalism and adaptability.

Fossil Record and what it tells us:

- Variety of fossils found in Africa, Asia and Europe, including almost full skeletons show:





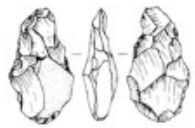
  - Shallow pelvis, long legs—efficient bipedal movement
  - Skull—increase in brain size to 1050 cubic cm, similar structure to modern humans.
  - Large brow ridge, thick facial bones, no defined chin, sloping forehead, broad, flat nose.
  - Tooth size decreasing—closer to modern humans
  - Foramen magnum at base of skull—bipedal
  - Limbs similar to modern humans

- Stone tools—more sophisticated than previous species. Fossil record for tools incomplete—may have used bamboo in Asia, which then degraded.

Cultural Evolution:

Tool Use

- Made hand-axes "Acheulian". Finely flaked, bifacial tools—hand axes.







- Used to butcher animals.

Fire Use

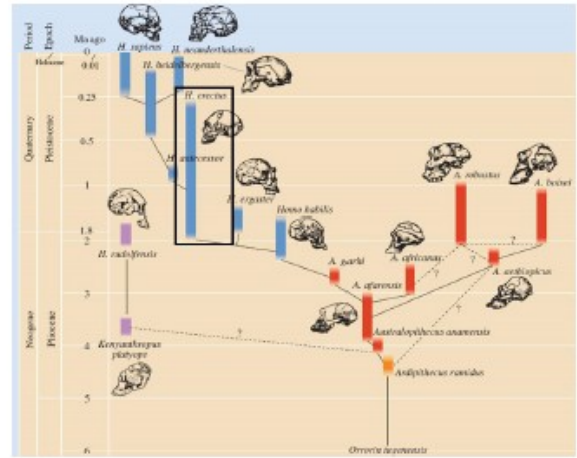
- Evidence of fire use in hunting, and cooking. May have had some control over fire.

Language, Art, Abstract Thought

- Likely to have had some language, but not as complex as modern humans
- No evidence of art or abstract thought.

Built Structures

- Made shelters from sticks and vegetation, also used caves.



Lifestyle:

- Modified environment to suit them
- Bipedal gait
- Family groups
- Made shelters and used tools
- Cared for elderly and disabled
- Planned hunting—eg driving elephants into swamps.

Diet and Dentition:

- Large amounts of meat, supplemented with plant foods.— Allowed for brain development
- Teeth show movement towards parabolic dental arcade:
  - Teeth smaller than previous species.

